

Subject Description Form

Subject Code	EE3007 / EE3007A
Subject Title	Computer System Principles
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: ENG2002
Objectives	1. To equip students with a comprehensive understanding of computer system organization and the internal architecture of a microprocessor 2. To enable students to understand software development for systems based on a SoC processor 3. To equip students with the ability to use a microprocessor or microcontroller to address engineering challenges
Subject Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Based on the specifications of an application, design the software and hardware to carry out the necessary operations based on a microprocessor or a microcontroller. b. Comprehend the advanced features of the latest microprocessors and the functions of basic computer peripherals. c. Understand the fundamentals of assembly language programming d. Develop logical thinking skills, analyze data effectively, and present results in written form.
Subject Synopsis/ Indicative Syllabus	Computer Systems Hardware and Operations 1. Microprocessor operations and its internal architecture: Operations of various registers, buses and data path, operations of ALU, arithmetic hardware, and general pipeline architecture. 2. Memory organization: Characteristics of memory technologies. Memory hierarchies and memory decoding mechanism. 3. Input and output systems: Direct I/O system and memory mapped I/O, interrupt and polling mechanisms. Typical serial data communication mechanisms. 4. Introduction to embedded computing systems: System organization and design of input/output system. Programming software for embedded systems. 5. Introduction to assembly language programming Laboratory Experiment: Perform basic input/output operations of an embedded system using the Python programming. Applications of different serial communication methods. Software development based on interrupt.
Teaching/Learning Methodology	Lectures and tutorials are the primary means of conveying the basic concepts and theories. Experiences in design, practical applications, and programming are achieved through experiments in which the students are expected to solve design problems with real-life constraints and to attain feasible solutions with critical and analytical thinking. Interactive laboratory sessions are introduced to encourage better preparation and hence understanding of the experiments. On-the-spot assessments are conducted in the laboratory to provide additional incentives for student's learning. Experiments are designed to supplement the lecturing materials, especially in Python programming, so that the students are encouraged to take extra readings and to look for relevant information.

	Teaching/Learning Methodology		Outcomes				
			a	b	c	d	
	Lectures		✓	✓	✓		
	Tutorials		✓	✓	✓		
	Experiments		✓		✓	✓	
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks		% weighting	Intended subject learning outcomes to be assessed			
				a	b	c	d
	1. Examination		60%	✓	✓	✓	✓
	2. Mid-term quiz		15%	✓	✓	✓	
	3. Laboratory performance & report		15%	✓			✓
	4. Online assignments and in-class activities		10%	✓		✓	✓
	Total		100%				
	It is a fundamental computer architecture subject. The outcomes on concepts, design, and applications are assessed by the usual means of examination and quiz, whilst those on analytical skills, problem-solving techniques, and practical considerations of programming, as well as technical reporting, are evaluated via experiments and the report.						
Student Study Effort Expected	Class contact:						
	▪ Lecture/Tutorial					30 Hrs.	
	▪ Laboratory					9 Hrs.	
	Other student study effort:						
	▪ Laboratory preparation/report					16 Hrs.	
	▪ Self-study					50 Hrs.	
	Total student study effort					105 Hrs.	
Reading List and References	Reference books and online materials:						
	1. J.L. Hennessy and D.A. Patterson, Computer Architecture: A Quantitative Approach, 6 th Edition, Elsevier, 2019						
	2. P. Darche, Microprocessor: Software and Hardware Aspects of Development, Debugging and Testing, John Wiley & Sons, 2021						
	3. C. Hamacher, Z. Vranesic. Et al., Computer Organization and Embedded Systems, McGraw Hill, 2022						
	4. G. Donzellini, A. M. Garavagno, et al., Introduction to Microprocessor-Based Systems Design, Springer, 2021						
	5. A.B. Downey, Think Python: How to Think Like a Computer Scientist, 2 nd ed., O'Reilly, 2015						
	6. S. Monk, Programming the Raspberry Pi Getting Started with Python, McGraw Hill, 2016						
	7. https://www.raspberrypi.org/documentation/usage/python/						